

Team / University: UKC AstraNova / University of Kent

Section	What Went Well?	Even Better If?
Quality of launch vehicle manufacture	Assembly process for parachute has been explained. Nice animation of both payloads and the battery being secured to the avionics sled.	It could have been explained how the motor mount attaches to the body tube, and how each component in the avionics sled will be secured. A video demonstration of the full assembly process (or some CAD images, could have communicated this.
Communication of manufacture	Materials and manufacturing methods have been identified for the main body tube, fins and nosecone.	These manufacturing method and material choices could have been explained and justified. For example, why the fin material was changed to MDF wood, why 3D-printed 'PLC' filament (presenter may have meant to say PCL or PLA) was chosen for the nosecone, etc.
Testing	Rideshare payload PCB components have been tested, and the results have been discussed.	Structural tests have been conducted for the main body and fins, but the results could have been discussed more, and supporting images could have been provided to show each test setup.
Media	Photos of the real launch vehicle parts and components have been presented. Clear and sufficiently edited video.	These photos could have been labelled and referred to more during discussions.
Quality of rideshare payload manufacture	Functional rideshare payload with a standard PCB design. Some of the PCB component selection was discussed, e.g. buzzer and switches.	More PCB components could have been discussed, e.g. which microcontroller is being used, which sensors, etc.

Systems engineering	The requirement compliance analysis table has been updated and discussed. Requirements are referenced throughout the presentation.	Some interfaces could have been defined in more detail. More evidence of iterative refinement could have been presented. It could have been explained how test results would have been/will be used to further refine the design, if necessary.
Project management	Financial budget has been broken down into individual costs/expenses. Mass and power budgets have also been provided.	Gantt chart could have shown more recent tasks, instead of just last November. Risks could have been presented in a risk assessment to assign mitigation measures and risk scores (= likelihood x impact) to each identified risk.
Simulation data	Insight provided for OpenRocket simulation and graphs. The launch vehicle's dynamic stability remains within the requirement margins.	Alternate designs could have been simulated. It would have been nice to see a comparison between CDR and MR designs. Additional graphs (beyond altitude and stability vs. time) could have been plotted.
General Feedback	The key thing the team could have elaborated further on was the launch vehicle's assembly process, i.e. how all the parts and components would come together. This was especially true for the motor mount and avionics sled, and how they would be attached to the body tube. Although these parts may have not been manufactured yet, some CAD images could have been used instead of real images. Otherwise, this was a good MR. We look forward to reviewing your FRR and preparing your team for the competition day in June!	

